MATH2601 – Higher Linear Algebra

Topic 2 – Vector spaces
Lecture 2.2 – Linear combinations and bases

Lecturer: Dr Sean Gardiner – sean.gardiner@unsw.edu.au

Linear combinations and span

Definition. Given a vector space $(V, \mathbb{F})$ and a set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} \subseteq V$, a linear combination of S is any expression of the form

$$\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n \text{ where } \lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{F}.$$

Definition. The span of a set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in a vector space $(V, \mathbb{F})$, denoted $\text{span}(S)$ (or $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$), is the set of all possible linear combinations of S. That is,

$$\text{span}(S) = \{\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n : \lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{F}\}.$$

Note that since the empty sum is canonically 0, we define $\text{span}(\varnothing) = \{\mathbf{0}\}$.

Lemma. For any vector space $(V, \mathbb{F})$ and set $S \subseteq V$, we have $\text{span}(S) \leq V$.

Proof. Clearly $S \subseteq \text{span}(S)$, so $\text{span}(S) \neq \varnothing$. Now let $\lambda \in \mathbb{F}$ and let $\mathbf{u}, \mathbf{v} \in \text{span}(S)$, so that $\mathbf{u}$ and $\mathbf{v}$ are linear combinations of elements of S. Then $\mathbf{u} + \lambda\mathbf{v}$ is also a linear combination of elements of S, and is therefore an element of $\text{span}(S)$. So by the subspace lemma, $\text{span}(S) \leq V$.

Notice that by construction, $\text{span}(S)$ is the smallest subspace of V containing S. In the case that $\text{span}(S) = V$, we say S is a spanning set for V or that S spans V.

Fact. If $(V, \mathbb{F})$ is a vector space and $S \subseteq T \subseteq V$, then $\text{span}(S) \leq \text{span}(T)$.

Examples of spanning sets

Example. The span of two non-parallel geometric vectors in $\mathbb{R}^n$ is a plane through the origin. The span of two parallel geometric vectors in $\mathbb{R}^n$ is a line through the origin.

Example. We have that $\mathcal{P}_n(\mathbb{R}) = \text{span}(\{1, t, t^2, \dots, t^n\})$ for any $n \in \mathbb{N}$. (We shall typically use t to represent the indeterminate variable in a polynomial vector space.)

Exercise. Let $\mathbf{u}, \mathbf{v}, \mathbf{b} \in \mathbb{R}^3$ with $\mathbf{u} = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$, and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$. Find conditions on $\mathbf{b}$ such that $\mathbf{b} \in \text{span}(\{\mathbf{u}, \mathbf{v}\})$.

Solution. We want to establish when there exist $\lambda_1, \lambda_2 \in \mathbb{R}$ such that $\lambda_1 \mathbf{u} + \lambda_2 \mathbf{v} = \mathbf{b}$. This yields a system of linear equations in λ_1 and λ_2 with corresponding augmented matrix

$$\left(\begin{array}{cc|c} 1 & 2 & b_1 \\ 4 & 3 & b_2 \\ 3 & 1 & b_3 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & -5 & b_2 - 4b_1 \\ 0 & 0 & b_1 - b_2 + b_3 \end{array}\right).$$

A row echelon form of this matrix shows the system will have no solutions unless $b_1 - b_2 + b_3 = 0$. So geometrically, $\mathbf{b}$ must lie in this plane.

Linear independence

Definition. Given a vector space $(V, \mathbb{F})$ containing a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subseteq V$, the vectors in S are called linearly independent if the only solution to

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_n \mathbf{v}_n = \mathbf{0} \quad \text{where} \quad \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$$

is $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$. (This is called the trivial solution.)

The vectors in S are linearly dependent if there is a non-trivial solution to the above equation – that is, if there is a solution to

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_n \mathbf{v}_n = \mathbf{0} \quad \text{where} \quad \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$$

with at least one of the scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ not equal to zero.

Example. The set $\{1, t+1, t^2+t+1\}$ is linearly independent in $\mathcal{P}_2(\mathbb{R})$. (This is because, writing $\lambda_1(1) + \lambda_2(t+1) + \lambda_3(t^2+t+1) = 0$ for $\lambda_i \in \mathbb{R}$, the t^2 coefficient implies $\lambda_3 = 0$, after which the t coefficient implies $\lambda_2 = 0$, after which the constant coefficient implies $\lambda_1 = 0$.)

Fact. Any subset of a linearly independent set is also linearly independent.

Linear independence example

Exercise. Do the vectors $p_1 = t^2 + 4t + 3$, $p_2 = 2t^2 + 3t + 1$, and $p_3 = t^2 - t$ form a linearly independent set in $\mathcal{P}_2(\mathbb{R})$?

Solution. We wish to find all solutions to the equation $\lambda_1 p_1 + \lambda_2 p_2 + \lambda_3 p_3 = 0$ for $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$. Equating coefficients of different powers of t yields a system of linear equations in λ_1, λ_2, and λ_3 with corresponding augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 4 & 3 & -1 & 0 \\ 3 & 1 & 0 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -5 & -5 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right).$$

A row echelon form of this matrix has leading entries in all columns except the last, meaning the system has a unique solution, which must be the trivial solution $\lambda_1 = \lambda_2 = \lambda_3 = 0$.

So $\{p_1, p_2, p_3\}$ is indeed a linearly independent set.

Reducing linearly dependent sets

Lemma. If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a set of vectors in a vector space $(V, \mathbb{F})$, then S is linearly dependent if and only if there is some element $\mathbf{v}_k \in S$ for which $\mathbf{v}_k \in \operatorname{span}(\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\})$.

Proof. For the forward direction, if S is linearly dependent, there is some nontrivial solution to $\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n = \mathbf{0}$. Let i be the greatest subscript for which λ_i is nonzero. Then $\lambda_i\mathbf{v}_i = \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_{i-1}\mathbf{v}_{i-1}$, and since λ_i^{-1} exists in $\mathbb{F}$, we can premultiply both sides of the equation by λ_i^{-1} to write $\mathbf{v}_i$ as a linear combination of the vectors that precede it in S:

$$\mathbf{v}_i = \lambda_i^{-1}\lambda_1\mathbf{v}_1 + \lambda_i^{-1}\lambda_2\mathbf{v}_2 + \cdots + \lambda_i^{-1}\lambda_{i-1}\mathbf{v}_{i-1} \in \operatorname{span}(\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{i-1}\}).$$

For the reverse direction, note that if $\mathbf{v}_k \in \operatorname{span}(\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\})$ then $\mathbf{v}_k = \mu_1\mathbf{v}_1 + \mu_2\mathbf{v}_2 + \cdots + \mu_{k-1}\mathbf{v}_{k-1}$ where not all of the μ_i are 0, meaning the equation $\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n = \mathbf{0}$ has the nontrivial solution $\lambda_i = \mu_i$ for $1 \le i < k$, $\lambda_k = -1$, and $\lambda_i = 0$ for $k < i \le n$.

Corollary. If S is a linearly dependent set in a vector space V, then there is some vector $\mathbf{v} \in S$ such that $\operatorname{span}(S - \{\mathbf{v}\}) = \operatorname{span}(S)$.

Corollary. If S is a linearly independent set in a vector space V and $\mathbf{v} \notin S$, then $\mathbf{v} \in \operatorname{span}(S)$ if and only if $S \cup \{\mathbf{v}\}$ is linearly dependent.

Sizes of linearly independent and spanning sets

Theorem. (Exchange lemma)
In any vector space $(V, \mathbb{F})$ with a finite spanning set, if $S \subseteq V$ is linearly independent and $T \subseteq V$ is a spanning set for V, then there is some spanning set $U \subseteq V$ with $|U| = |T|$ for which $S \subseteq U$ and $U - S \subseteq T$.

Proof. Let S be linearly independent and T be spanning in V, with $|S| = m$ and $|T| = n$ for some $m, n \in \mathbb{N}$. Let $P(k)$ be the statement "k elements of S can be swapped into T without changing its span". Clearly $P(0)$ is true. Suppose now that $P(k)$ is true for some $0 \leq k < \min(m, n)$, so that the set $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k, \mathbf{t}_{k+1}, \dots, \mathbf{t}_n\}$ spans V and each $\mathbf{s}_i \in S$ and $\mathbf{t}_i \in T$. Since $\mathbf{s}_{k+1}$ lies in the span of this set, from the previous lemma we know that the set $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k, \mathbf{s}_{k+1}, \mathbf{t}_{k+1}, \dots, \mathbf{t}_n\}$ is linearly dependent, and that there is some vector $\mathbf{v}$ in the set that is a linear combination of the vectors preceding it. Furthermore, this vector $\mathbf{v}$ cannot be one of the $\mathbf{s}_i$, otherwise the previous lemma would imply $\{\mathbf{s}_1, \dots, \mathbf{s}_i\}$ is linearly dependent, contradicting the fact any subset of a linearly independent set is linearly independent. So the vector $\mathbf{v}$ is one of the $\mathbf{t}_i$ vectors, and we can safely remove it from the set without changing its span. The resultant set has $k + 1$ elements of S, showing that $P(k + 1)$ is true whenever $P(k)$ is true. So the statement is true by induction.

Sizes of linearly independent and spanning sets

Theorem. (Exchange lemma)
In any vector space $(V, \mathbb{F})$ with a finite spanning set, if $S \subseteq V$ is linearly independent and $T \subseteq V$ is a spanning set for V, then there is some spanning set $U \subseteq V$ with $|U| = |T|$ for which $S \subseteq U$ and $U - S \subseteq T$.

Proof (cont'd). We have shown that up to $\min(m, n)$ elements in S can be swapped into T without changing its span. Notice that if $m \leq n$, performing these swaps m times will create the desired set U for which $|U| = |T|$, $S \subseteq U$, and $U - S \subseteq T$.

It remains to deal with the $m > n$ case. Suppose that $m > n$ and that we have swapped all elements of T out, so that $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_n\}$ spans V. But then $\mathbf{s}_{n+1}$ lies in the span of this set, implying $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_n, \mathbf{s}_{n+1}\}$ is linearly dependent. This is a contradiction since $\{\mathbf{s}_1, \dots, \mathbf{s}_{n+1}\}$ is a subset of a linearly independent set. So it is not possible for m to exceed n.

Corollary. In any vector space $(V, \mathbb{F})$ with a finite spanning set, the size of any linearly independent set is less than or equal to the size of any spanning set. That is, if $S \subseteq V$ is linearly independent and $T \subseteq V$ is a spanning set for V, then $|S| \leq |T|$.

Bases

Definition. Given a vector space V, a basis for V is any set of vectors $S \subseteq V$ for which both $\text{span}(S) = V$ and S is linearly independent.

Example. We are already familiar with many standard bases for certain types of vector spaces:

- The standard basis for $\mathbb{F}^n$ is $B = \{\mathbf{e}_i : 1 \leq i \leq n\}$ where each $\mathbf{e}_i$ is the column vector with ith component 1 and all other components 0. (In $\mathbb{R}^3$ these are often called $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.)
- The standard basis for $M_{p,q}(\mathbb{F})$ is $B = \{E_{i,j} : 1 \leq i \leq p, 1 \leq j \leq q\}$ where each $E_{i,j}$ is the $p \times q$ matrix with (i,j)th component 1 and all other components 0.
- The standard basis for $\mathcal{P}_n(\mathbb{F})$ is $B = \{1, t, t^2, \dots, t^n\}$.
- For any finite nonempty set $X = \{a_1, a_2, \dots, a_n\}$, the standard basis for $\mathcal{F}(X)$ is $B = \{f_i : X \to \mathbb{F} : 1 \leq i \leq n\}$ where $f_i(a_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$
- For the vector space $(\mathbb{C}, \mathbb{R})$, the standard basis is $B = \{1, i\}$.
- For the vector space $(\mathbb{Q}(\sqrt{2}), \mathbb{Q})$, the standard basis is $B = \{1, \sqrt{2}\}$.

Dimension

Lemma. If a vector space V has a finite basis, then every basis of V has the same size.

Proof. Suppose we have two bases B_1 and B_2 of V. Since B_1 is linearly independent and B_2 is spanning, $|B_1| \leq |B_2|$. But since B_2 is linearly independent and B_1 is spanning, $|B_2| \leq |B_1|$. So $|B_1| = |B_2|$.

Definition. If a vector space $(V, \mathbb{F})$ has a finite basis, we call the size of the basis the dimension of the vector space, written $\dim(V)$ or $\dim_{\mathbb{F}}(V)$. Otherwise, we say the vector space has infinite dimension.

Exercise. Find the following vector space dimensions:

- $\dim(M_{p,q}(\mathbb{F})) = pq$.
- $\dim(\mathcal{P}_n(\mathbb{F})) = n + 1$.
- $\dim_{\mathbb{R}}(\mathbb{C}) = 2$.
- $\dim_{\mathbb{C}}(\mathbb{C}) = 1$.
- $\dim(\mathcal{P}(\mathbb{F})) = \infty$.
- $\dim(\{\mathbf{0}\}) = 0$.

Fact. If V is a finite-dimensional vector space and $S \subseteq V$, then S is a basis for V if and only if any two of the following three properties hold:

- S spans V.
- S is linearly independent.
- $|S| = \dim(V)$.

Dimension example

Exercise. Find the dimension of the vector space $(V, \mathbb{R})$ whose elements are all symmetric 2×2 matrices over $\mathbb{R}$.

Solution. A 2×2 symmetric matrix has the form $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ for some $a, b, c \in \mathbb{R}$. Since this can be decomposed as

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

we can see $B = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ is a spanning set for V.

Furthermore, since B is clearly linearly independent, B is a basis for V. So $\dim(V) = |B| = 3$.

Exercise. Show that for any $n \in \mathbb{Z}^+$, the vector space of all symmetric $n \times n$ matrices over $\mathbb{R}$ has dimension $\frac{1}{2}n(n+1)$.

Building bases

Theorem. If S is a spanning set for a finite-dimensional vector space V, then $B \subseteq S$ for some basis B of V.

Proof. Let S be a spanning set in V. Recall from the first corollary on slide 5 that if S is linearly dependent, it is possible to remove a vector from the set without changing its span. This vector removal process can be repeated until the remaining set is linearly independent, and still spans V, at which point the resultant set is a basis for V.

Corollary. If a vector space V contains a finite spanning set, then it has finite dimension.

Theorem. If S is a linearly independent set in a finite-dimensional vector space V, then $S \subseteq B$ for some basis B of V.

Proof. Let U be a (finite) spanning set for V that contains S, which by the exchange lemma must exist. Consider all linearly independent subsets of U that contain S, and let B be one such subset that is of maximal size. Then any larger subset is linearly dependent, so for every $\mathbf{v} \in U - B$, we have $\mathbf{v} \in \text{span}(B)$, and of course for every $\mathbf{v} \in B$ we also have $\mathbf{v} \in \text{span}(B)$. So $\text{span}(B) = \text{span}(U) = V$, meaning B is spanning and thus is a basis for V.

Unique representations

Theorem. Given a finite-dimensional vector space $(V, \mathbb{F})$, the set $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for V if and only if every $\mathbf{v} \in V$ can be written uniquely as $\mathbf{v} = \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n$ for some $\lambda_i \in \mathbb{F}$.

Proof. For the forward direction, suppose B is a basis for V. Then since it spans V, certainly every vector in V can be written as a linear combination of B. Suppose that there are two linear combinations for some particular $\mathbf{v} \in V$ such that

$$\mathbf{v} = \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n = \mu_1\mathbf{v}_1 + \mu_2\mathbf{v}_2 + \cdots + \mu_n\mathbf{v}_n$$

for some $\lambda_i, \mu_i \in \mathbb{F}$. Then

$$\mathbf{0} = (\lambda_1 - \mu_1)\mathbf{v}_1 + (\lambda_2 - \mu_2)\mathbf{v}_2 + \cdots + (\lambda_n - \mu_n)\mathbf{v}_n,$$

which by the linear independence of B has the unique solution $\lambda_i - \mu_i = 0$, so $\lambda_i = \mu_i$ for all $1 \le i \le n$. So there is a unique linear combination for each $\mathbf{v} \in V$.

For the reverse direction, suppose every $\mathbf{v} \in V$ can be written uniquely as a linear combination of B. Then B clearly spans V, and since in particular there is a unique linear combination for $\mathbf{0}$, we know B is linearly independent. So B is a basis for V.

Coordinate vectors

Definition. Given a finite-dimensional vector space $(V, \mathbb{F})$ with ordered basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$, and a particular vector $\mathbf{v} \in V$, the unique scalars $\lambda_i \in \mathbb{F}$ for which $\mathbf{v} = \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n$ are called the coordinates of $\mathbf{v}$ with respect to B, and we define the coordinate vector of $\mathbf{v}$ with respect to B to be $[\mathbf{v}]_B = (\lambda_i)_{0 \le i \le n}$ (an element of $\mathbb{F}^n$).

Fact. The following statements are true for any vector space $(V, \mathbb{F})$ with any basis B. (These are easy to prove as an exercise.)

- For all $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} = \mathbf{v}$ if and only if $[\mathbf{u}]_B = [\mathbf{v}]_B$.
- For all $\mathbf{u}, \mathbf{v} \in V$, $[\mathbf{u} + \mathbf{v}]_B = [\mathbf{u}]_B + [\mathbf{v}]_B$.
- For all $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$, $[\lambda\mathbf{v}]_B = \lambda[\mathbf{v}]_B$.

This means that every finite-dimensional vector space $(V, \mathbb{F})$ can be faithfully mapped into a column vector space $\mathbb{F}^n$ where $n = \dim(V)$.

Example. For each of the standard bases listed earlier on slide 8, the corresponding coordinate vectors are particularly nice. For example, in $\mathcal{P}_n(\mathbb{F})$, using the standard ordered basis $B = \{1, t, t^2, \ldots, t^n\}$ we have

$$[a_0 + a_1 t + \cdots + a_n t^n]_B = (a_i)_{0 \le i \le n} \in \mathbb{F}^{n+1}.$$

Coordinate vectors example

Exercise. In the vector space $\mathcal{P}_2(\mathbb{R})$, find the coordinates of $9t^2 + 11t + 10$ with respect to the ordered basis $B = \{t^2 + 4t + 3,\ 2t^2 + 3t + 1,\ t^2 - t\}$.

Solution. We want to find $\lambda_i \in \mathbb{R}$ such that

$$\lambda_1(t^2 + 4t + 3) + \lambda_2(2t^2 + 3t + 1) + \lambda_3(t^2 - t) = 9t^2 + 11t + 10.$$

Equating coefficients of different powers of t yields a system of linear equations in λ_1, λ_2, and λ_3 with corresponding augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 9 \\ 4 & 3 & -1 & 11 \\ 3 & 1 & 0 & 10 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 9 \\ 0 & -5 & -5 & -25 \\ 0 & 0 & 2 & 8 \end{array}\right).$$

Solving via back-substitution gives the solution $\lambda_3 = 4$, $\lambda_2 = 1$, and $\lambda_1 = 3$.

So we have $[9t^2 + 11t + 10]_B = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$.

Note that in this case, we could have quickly solved for λ_3 by substituting $t = -1$ into the first equation, yielding $\lambda_1(0) + \lambda_2(0) + \lambda_3(2) = 8$. (Other substitutions like $t = 0$ and $t = 1$ can produce other nice expressions that make the problem-solving slightly easier.)